



Semester 1
Examinations 2025-2026

Course Instance Code(s)	1CSD1, 1MAI1, 1OA2, 2MF1, 4BCT1, 4BS2
Exam(s)	M.Sc. in Computer Science (Data Analytics), M.Sc. in Computer Science (Artificial Intelligence), Occasional Arts, M.Sc. in Software Design and Development, B.Sc. in Computer Science and Information Technology B.Sc. in Science
Module Code(s)	CT4100
Module(s)	Information Retrieval
Bonded	No ☒] Yes [] List bonds
Paper No.	1
External Examiner(s)	Dr. Shoaib Ehsan
Internal Examiner(s)	Dr. Enda Howley *Dr. Colm O'Riordan

Instructions: Answer question 1 and any two other questions.
Duration 2 hours
No. of Pages 3
Discipline(s) English

Requirements:

Take from exam venue, release online	No []	Yes ☒
MCQ Answer sheet	No ☒	Yes []
Log Tables	No ☒	Yes []
Cambridge Tables 2 nd Edition	No ☒	Yes []
Graph Paper	No ☒	Yes []
Graphic material in colour	No ☒	Yes []

End of requirements.

PTO

CT4100 Information Retrieval

Exam Duration: 2 Hours

You must complete Question 1 (40 marks) and any two other questions (20 marks each).

Question 1

- (a) Describe the vector space model and explain how similarity is defined between a document and a query using the vector space model. (5 marks)
- (b) What is meant by the *term independence* assumption? Explain how this assumption may impact the performance of IR systems and suggest a means to overcome the assumption. (5 marks)
- (c) In addition to the metrics of precision and recall, the diversity of documents in the answer set is also of importance to users. Suggest an approach to measure the diversity of the top-k documents returned. (5 marks)
- (d) Explain why two identical documents concatenated together would have a higher similarity score than the original document when using basic *tf-idf* term weighting without normalisation. What problem does this highlight, and how do modern weighting schemes address it? (5 marks)
- (e) Explain the main components of the BM25 weighting scheme. What role do the two tuning parameters play? (5 marks)
- (f) The Rocchio method for relevance feedback relies on binary relevance feedback. Suggest how one could augment the approach to allow a range of weights in feedback? (5 marks)
- (g) Relevance-feedback relies on the assumption that relevant documents cluster together in term-vector space. Under what circumstances might this assumption fail, and how would this affect the effectiveness of feedback techniques? (5 marks)
- (h) Explain what is meant by Mean Average Precision (MAP). What are the benefits and limitations of using MAP as an evaluation metric. (5 marks)

Question 2

(a) Query performance prediction aims to estimate retrieval effectiveness before examining relevance judgments by predicting query difficulty. Describe methods, with examples, for predicting query difficulty based on:

- i) Query-specific characteristics and features.
- ii) Statistical properties of the document collection.
- iii) Analysis of the retrieved result set.

(10 marks)

(b) Describe a collaborative filtering approach that can be used in an online movie recommendation system. Explain how you would handle the cold start problem when new users join the platform with no rating history. How could such a system automatically identify users for whom it may be difficult to provide recommendations? (10 marks)

Question 3

(a) Query expansion aims to improve recall by adding related terms to the user's original query. Compare and contrast two different approaches to query expansion: one based on local analysis of the return set, and one based on a fixed thesaurus or ontology.

(10 marks)

(b) HITS (Hyperlink-Induced Topic Search) is an alternative link-based ranking algorithm that identifies hub and authority pages. Explain the concepts of hubs and authorities, and describe how the HITS algorithm computes these scores. How does HITS differ from PageRank in its approach and application? (10 marks)

Question 4

(a) Learning to Rank (LTR) applies machine learning approaches to automatically learn ranking functions from training data. Describe the key advantage of LTR approaches compared to manually-tuned ranking functions. Outline what constitutes a training example and what evaluation metrics you could use to guide the search. (10 marks)

(b) Neural learning to rank models have shown strong performance in recent years. Explain how neural LTR models differ from traditional LTR approaches in terms of:

- Feature representation.
- Computational requirements.
- Interpretability.

(10 marks)

END